

Lab 23: Extending Existing Classes

- Due Friday by 11:59pm
- Points 5
- Submitting an external tool
- Available Jul 22 at 6pm - Jul 24 at 11:59pm

Learning Objectives

- Define custom exceptions
- Use custom exceptions
- Use class attributes.

Completion Requirements

- Submit Lab23.py

Q1: Periodic Table of Elements

In this question, you will model chemical elements and reactions using inheritance, custom exception classes, list copying, and class-level attributes. All classes must be defined in `Lab23.py`.

Custom Exception Classes

Define four custom exception classes that inherit directly from Python's built-in `Exception` class:

- `InvalidProtonNumberError`
- `InvalidNeutronNumberError`
- `UndiscoveredElementError`
- `UndefinedReactionError`

Element (Base Class)

Attributes



- `name` (str): The name of the element (e.g., `"Sodium"`).
- `protons` (int): The atomic number representing proton count.
- `neutrons` (list of int): A copy of the list of neutron counts for various isotopes.

Methods

- **__init__(self, name, protons, neutrons):**

Before assigning attributes, perform input validation in the following order:

1. If `protons < 1`, raise `InvalidProtonNumberError` with the message: "Proton number must be greater than 1"
2. If `protons > 118`, raise `UndiscoveredElementError` with the message: "Proton number is higher than the maximum element discovered so far"
3. If any integer in the `neutrons` list is negative (< 0), raise `InvalidNeutronNumberError` with the message: "Number of neutrons can not be negative"

After passing all validation checks, assign `self.name`, `self.protons`, and assign a **copy** of the input `neutrons` list to `self.neutrons` to prevent external modification.

- **getIsotopes(self):**

Calculates and returns a list of strings representing isotope names. Each isotope string is created by combining the element's name, a hyphen, and the mass number (sum of protons and neutrons for that isotope).

Formula: `f"{self.name}-{(self.protons + n)}"` for each neutron count `n` in `self.neutrons`.

Group1Element (Derived Class)

Create a class `Group1Element` that inherits directly from `Element`.

Class Attributes

- `reactions` (dict): A class-level dictionary defining alkali metal reactions:

```
reactions = {'water': 'hydroxide', 'oxygen': 'oxide'}
```

Methods

- **__init__(self, name, protons, neutrons):**

Accepts the three standard parameters and passes them directly to `super().__init__(name, protons, neutrons)`. The parent class will handle all attribute assignments and input validation.

- **react(self, reactant):**

Looks up the string `reactant` in the class attribute dictionary `Group1Element.reactions`.

- If the reactant exists in the dictionary, combine the element's `name`, a space, and the retrieved product suffix, then return the string (e.g., "Potassium hydroxide").
- If the reactant is not found in `reactions`, raise an `UndefinedReactionError` with the exact message: `f"Reaction with {reactant} is not defined"`.



Examples (Normal Cases)

```
>> Na = Element("Sodium", 11, [11, 12, 13])
>> Na.getIsotopes()
['Sodium-22', 'Sodium-23', 'Sodium-24']
```

```
>> K = Group1Element('Potassium', 19, [20, 21, 22])
>> K.getIsotopes()
['Potassium-39', 'Potassium-40', 'Potassium-41']

>> K.react('water')
'Potassium hydroxide'
```

Examples (Exception Cases)

```
>> K = Group1Element('Potassium', -1, [20, 21, 22])
InvalidProtonNumberError: Proton number cannot be negative

>> K = Group1Element('Potassium', 120, [20, 21, 22])
UndiscoveredElementError: Proton number is higher than the maximum element discovered so far

>> K = Group1Element('Potassium', 19, [20, 21, -22])
InvalidNeutronNumberError: Number of neutrons can not be negative

>> K = Group1Element('Potassium', 120, [20, 21, -22])
UndiscoveredElementError: Proton number is higher than the maximum element discovered so far

>> K = Group1Element('Potassium', 19, [20, 21, 22])
>> K.react('chlorine')
UndefinedReactionError: Reaction with chlorine is not defined
```

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